Designer Genes – 2026 Greenbrier Invitational - Division C

Section 1: Multiple Choice (1 point each)

1. During DNA replication, which enzyme unwinds the DNA molecule?
    a. DNA polymerase
    b. Ligase
    c. Helicase
    d. Topoisomerase
    e. RNA polymerase

2. Which cellular structure is directly responsible for translation?
    a. Ribosome
    b. Nucleus
    c. Golgi Apparatus
    d. Smooth endoplasmic reticulum
    e. Mitochondria

3. Which of the following is NOT a difference between DNA and RNA?
    a. DNA is double stranded, RNA is not
    b. DNA contains adenine, RNA does not
    c. DNA has deoxyribose sugar, RNA does not

4. What is the most likely combination of sex chromosomes in a person who has Turner's Syndrome?
    a. XXY
    b. XO
    c. XXYY
    d. XY
    e. XYY

5. Which of the following results in Trisomy-21?
    a. Insertion point mutation
    b. Deletion point mutation
    c. Substitution point mutation
    d. Nondisjunction
    e. Nonsense mutation

6. A gene at one location that masks the expression of another gene is called
    a. Epistasis
    b. Chiasma
    c. Tetrasomy
    d. Nullisomy
    e. Pleiotropy

7. In eukaryotic cells, during which phase of the cell cycle is DNA replicated?
    a. Prophase
    b. Interphase - G1
    c. Telophase
    d. Interphase - S
    e. Interphase - G2

8. What does tRNA stand for?
    a. Translation Ribonucleic acid
    b. Transform Ribonucleic acid
    c. Translocation Ribonucleic acid
    d. Transfer Ribonucleic acid

9. During which phase of meiosis are sister chromatids pulled apart?
    a. Prophase I
    b. Prophase 2
    c. Anaphase 1
    d. Anaphase 2

10. The nitrogenous bases whose structure is has a double aromatic ring are:
    a. Purines
    b. Pyrimidines
    c. Cytosines
    d. Thymines

11. A somatic human cell typically has ____ pairs of chromosomes.
    a. 49
    b. 26
    c. 43
    d. 23

12. Which of Mendel's laws state that chromosomes do not influence each other during the sorting of chromosomes into the produced gametes during meiosis?
    a. Law of Segregation
    b. Law of Dominance
    c. Law of Independent Assortment
    d. Law of Inheritance

13. Which is NOT an example of a polygenic trait in humans?
    a. Height
    b. Skin color
    c. Hair color
    d. Color blindness

14. During which phase of meiosis does the nuclear membrane reform?
    a. Prophase I
    b. Anaphase I
    c. Prophase II
    d. Telophase II

15. What are the short lengths of DNA that are produced by the discontinuous replication of the lagging strand?
    a. Replication forks
    b. Okazaki fragments
    c. RNA polymerase
    d. Exons

16. Recessive X-linked disorders in humans occur:
    a. Predominantly in females
    b. Predominantly in males
    c. Both females and males equally
    d. Randomly through males and female populations

17. The crossing over event will most likely occur during which phase?
    a. Meiosis I
    b. Meiosis II
    c. Mitosis I
    d. Interphase

18. What makes up the backbone of DNA strands?
    a. Alternating phosphate groups and nitrogenous bases
    b. Alternating ribose and deoxyribose sugars
    c. Alternating deoxyribose sugar and phosphate groups
    d. Alternating nitrogenous bases and deoxyribose sugar

19. Which of the following terms are produced from meiosis?
    a. Somatic cells
    b. Gametes
    c. Diploid
    d. Autosomal

20. Where does transcription occur in a eukaryotic cell?
    a. Ribosome
    b. Mitochondria
    c. Golgi apparatus
    d. Nucleus
    e. Plasma membrane

Section 2: Short Answer (various points)

21. (4 pts) Unicorns have 2 alleles for horn color, which following Mendelian genetics. Blue horns (*B)* are dominant over white horns (*b*).

    a) What is the homozygous dominant genotype?
    b) What is the heterozygous genotype?
    c) What phenotype will result from a heterozygous genotype?
    d) What phenotype will result from a homozygous recessive genotype?

22. (3 pts) Write the complementary strand for the following DNA sequence.

    CCT-AGT-GAA

23. (13 pts) In a species of flower, a flower with red petals (*R*) is crossed with a flower with white petals (*W*). The resulting offspring have pink petals (*RW*).

    a) What kind of non-Mendelian inheritance is this?

    b) A pink flower is crossed with a red flower. Identify the genotypes of both parent organisms in this cross.

    c) Correctly create a Punnett square and provide the <u>percentages</u> of all expected genotypes and phenotypes on the provided answer sheet.

24. (1 pt) A sample of DNA consists of 10% adenine. What percentage of cytosine will it contain?

25. (13 points) Can a man with type A blood and a woman with type B blood have a child with type O blood?
    a) Yes, or no?

    b) Prove with a Punnett Square on the provided answer sheet.

26. (1 pt) What kind of bond(s) are found connecting the nitrogenous bases between the double strands of DNA?

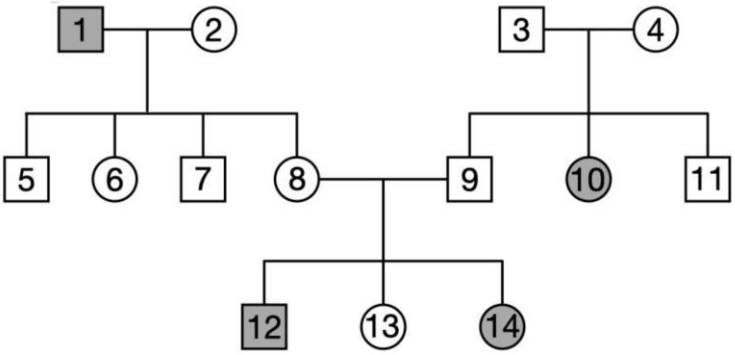
27. (8 pts) Given the following sequence of DNA: TAC-GTT-AAA-GUA

    a) What is the sequence of mRNA using the template strand?

b) What is the sequence of amino acids produced from the template strand, you may use the three letter abbreviation for the amino acid?

28. (3 pts) List all the possible “stop” codons for all known life?

29. (4 pts) The following table shows three generations of guinea pigs. In guinea pigs, rough coat (*R*) is dominant over smooth coat (*r*), and this trait follows autosomal Mendelian inheritance. Shaded individuals have smooth coats. Answer the following questions assuming that all information i



a) How many male children belong to individual #9?

b) What is the genotype of individual #2?

c) What is the phenotype of individual #10?

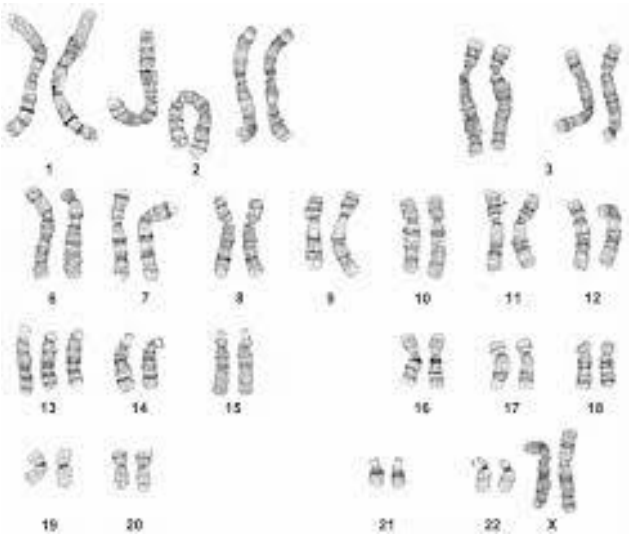
d) How many sisters does individual #10 have?

30. (2 pts) What cellular structure pulls the spindle fibers during mitosis and meiosis to move the chromosomes through the cytoplasm?

31. (3 pts) What does mRNA stand for (3 words total)?

32. (15 pts) Rabbits have multiple alleles for fur color on autosomal chromosomes. Full color (*R*) is dominant to chinchilla (r^{ch}) which is dominant to himalayan (r^{h}) which is dominant to albino (*r*). If a female full-color rabbit that had a mother with albinism is crossed with a male who is heterozygous for the himalayan coat.
a) What are the genotypes of the mother and father in the cross?
b) What will be the <u>percentages</u> of genotypes and phenotypes of their offspring? Show your work using a Punnett Square.

33. (6 pts) Use the karyotype below to answer the questions.



a) What is the most likely sex of this *Homo sapien* individual?

b) This individual most likely has which specific disorder?

c) On your answer sheet, circle all symptoms that will most likely occur with this disorder.

34. (2 pts) Lysine, Arginine, and Glutamine are examples of which biological monomer _____.

35. (2 pts) What does the “G” stand for in G1 and G2 of interphase?

36. (2 pts) a) Is blood type a mendelian or non-mendelian trait in humans?
b) Explain your answers in a short sentence.

37. (17 pts) In dragons, having five claws (*D*) is dominant over having four claws (*d*), and is an autosomal trait that follows Mendelian genetics. The color of scales is autosomal and exhibits codominance. The scales can be green (*G*) or purple (*P*), but heterozygotes show both scale colors together in a striped pattern (GP*)*. Answer the following questions for a cross between a male striped dragon with four claws and a female striped dragon who is heterozygous for five claws.

    a) What are the genotypes of the parent organisms in this cross?
    b) On the answer sheet, create a Punnett Square showing the expected offspring of this cross.
    c) What percentage of their offspring will most likely be green?
    d) What percentage of their offspring will most likely have the same phenotype as their father?
    e) What percentage of their offspring will most likely be heterozygous for both traits?

38. (2 pts) What is the difference between the process of transcription and translation in prokaryotes that makes them much faster at responding to stimuli than eukaryotes?

39. (2 pts) What is the name of the region of DNA upstream of a gene that often controls the gene's expression by controlling the binding of RNA polymerase?

40. (5 pts) During RNA processing in eukaryotic cells, the (a)_____ are removed from the pre-mRNA sequence and the (b)_____ are reconnected during RNA (c)_____. Then, the 5'-(d)_____ and the 3'-(e)_____ are added to make the mature mRNA strands.

41. (2 pts) What is the wobble effect in protein translation. Explain in less than 2 sentences.

42. (2 pts) A bacterial organism has a Trp operon within its genome. Tryptophan is easily available within the environment of this organism, what occurs to the operon?

43. (3 pts) A bacterial organism contains a Lac operon within its genome.

    a) What substance has to be available for a bacterial organism to induce a Lac operon?

    b) What substance will be produced once the Lac operon is induced?

    c) Why would this system be beneficial to this organism?

44. (2 pts) The mitochondrial DNA in all human cells most likely came from which parent?

45. (2 pts) During the creation of genetically modified organisms, what are used to cut the DNA strand at very specific locations to help control the insertion/deletion of the studied gene(s)?

46. (2 pts) What laboratory assay will take a DNA sample, cut it into many differently sized fragments, then run them through a semi-solid material to separate and visualize the strands?

47. (2 pts) The Central Dogma of Biology states that during transcription, DNA will always be converted into RNA. There is one famous viral agent that is an exception to this rule.

    a) What is the name of the viral agent?

    b) What is the name of the enzyme that allows for the ability to go from RNA to DNA in this viral family?

48. (2 pts) During gel electrophoresis of a DNA sample, the electricity will separate and move the DNA fragments by which two properties?